Applied Industrial IoT - Project

Solar Plant Manufacturing

BATCH NO: 12

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AIM:

To monitor and control solar power plant by using iot with the help of SCADA.

PROBLEM STATEMENT AND SOLUTION:

Solar power plant is used to generate high rate of power at less area, So there we want to monitor each and every thing at ever minute like weather, output power, inventor & transformer status.

And here we use Supervisory control and data acquisition (**SCADA**) system for control and monitor our inventors, sensors and on/off purpose. And other data also monitor in SCADA.

PROJECT DESIGN SPECIFICATIONS:

Weather:

In solar power plant we want to monitor weather regularly by using weather monitoring system.

Temperature monitoring:

Temperature sensor is used to sensing the temperature of Transformer.

Energy consumption:

Sometime the output of power plant will be decrease due to bad maintenance of connecting cables and solar panels, so we monitoring the plant continuously and we take immediate action to improve output.

Data monitoring and processing:

Supervisory Control and Data Acquisition (SCADA) is used for controlling, monitoring and processing the dates.

Communication protocol:

RS-485 Modbus allows multiple devices (up to 32) to communicate, so we use this Modbus.

Integrated of data to an iot platform:

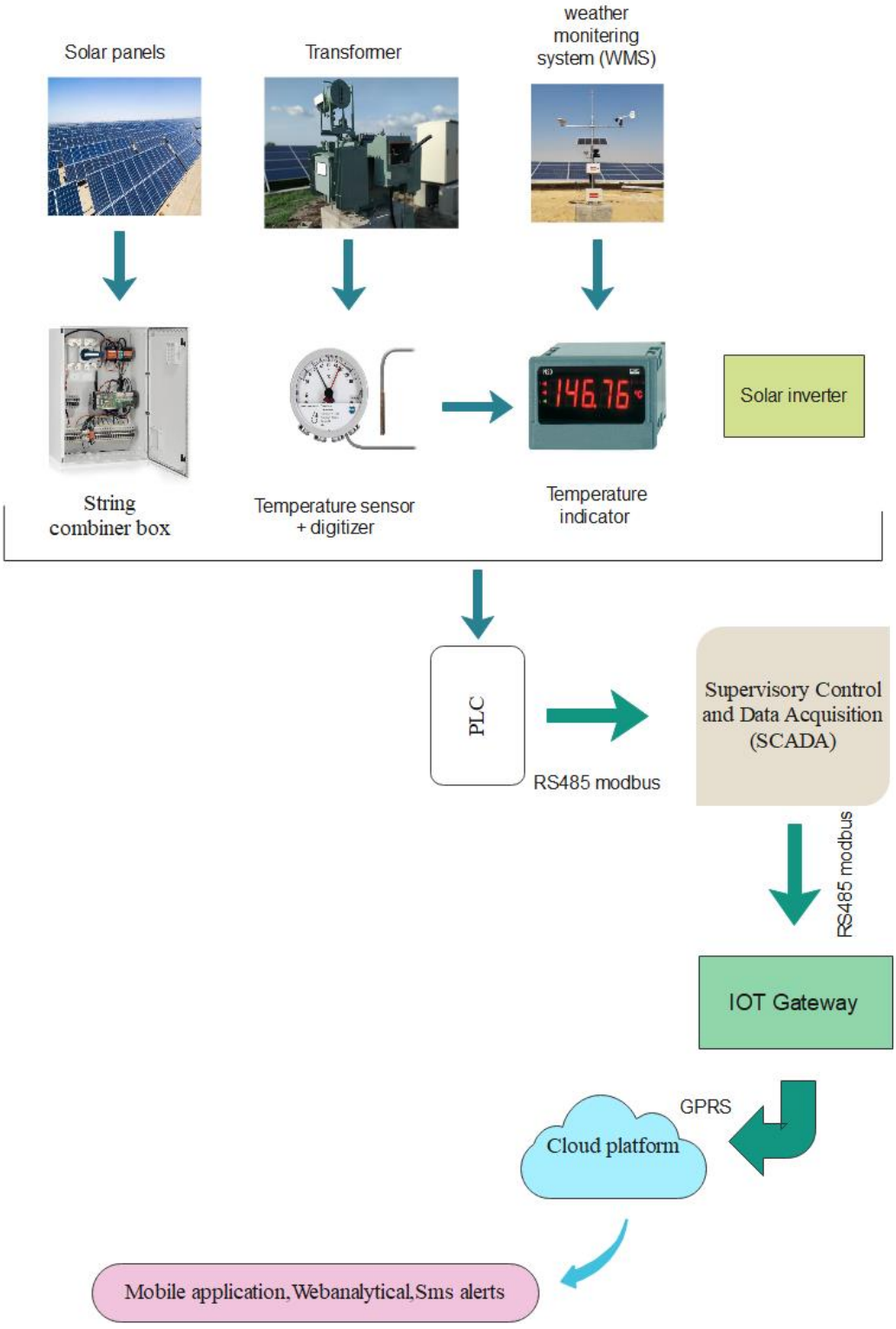
All the data collected from the above parameters are integrated to an IoT platform deployed on local or a cloud server. The IoT platform translates the data to information like graph, trends, alerts and other business intelligence reports.

Flow explanation:

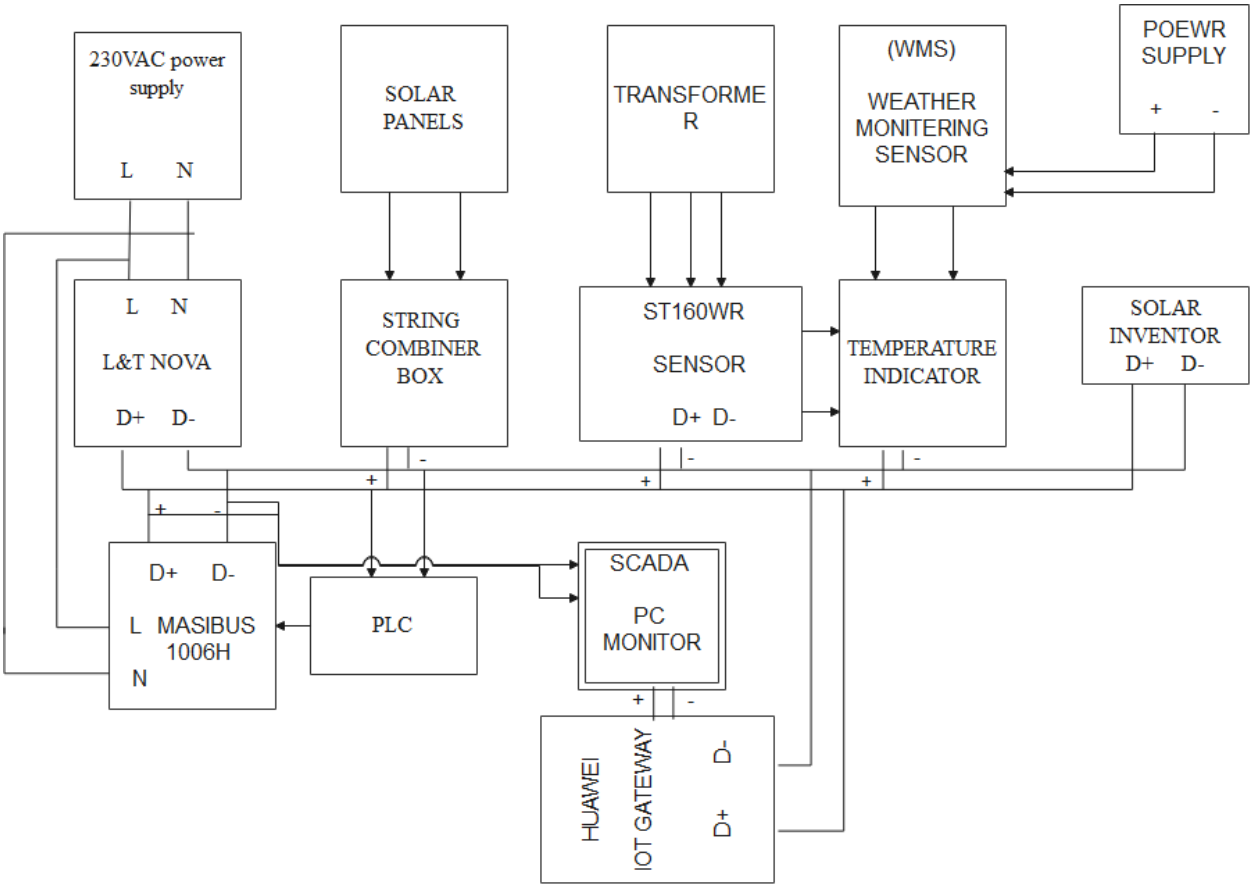
In solar power plant, solar panels are connected to solar string combiner box. And Transformer are connected with temperature sensor for monitoring transformer temperature, and temperature sensor connected to digitizer. We use weather monitoring sensor (WMS) for monitoring atmosphere temperature that WMS connected with temperature indicator. And we use solar inventor for converting AC to DC.

Above mention parts are connected to PLC by wires, and the PLC send the inputs to SCADA through Modbus. And SCADA will control and monitor all the process and SCADA send the input to IOT gateway through RS-485 Modbus, And the IOT gateway prepare data and send to cloud platform. When cloud receive data then the user get notification, alarm, sms at their system and mobile.

PROJECT ARCHITECTURE:



WIRING DIAGRAM



COMPONENTS WORKING PRINCIPLES / FUNCTIONALITY:

WEATHER MONITORING SYSTEM (WMS):

We have lot of Advance Weather Monitoring System use to measures solar radiation, wind speed, wind direction, PV temperature, ambient temperature, humidity and rainfall along with inverter and energy meter data. Weather monitoring station communicates with SCADA systems on RS485 MODBUS.

STIRNG COMBINER BOX:

The role of the combiner box is to bring the output of several solar strings together. Daniel Sherwood, director of product management at Solar BOS, explained that each string conductor lands on a fuse terminal and the output of the fused inputs are combined onto a single conductor that connects the box to the inverter.

TEMPERATURE SENSOR:

A temperature sensor is an electronic device that measures the temperature of transformer and converts the input data into electronic data through temperature indicator.

TEMPERATURE INDICATOR:

Temperature indicators are installation instruments which can process signals from temperature sensors and show them on the display. Temperature indicators enable easy and economic valuation of resistance sensors, such as Pt100 or different thermo element types.

SOLAR INVERTER:

A solar inverter is one of the most important elements of the solar electric power system. It converts the variable direct current (DC) output of a photovoltaic (PV) solar panel into alternating 240V current (AC). This AC electricity then can be fed into your home to operate your appliances.

DIGITIZER:

A digitizer is an electronic device that converts an analogue signal into a digital format and provides the corresponding output in terms of display or on its communication port. Inputs to digitizer will be analogue signal from sensors like temperature, pressure, flow etc.

Programmable logic controller: (PLC)

PLC stands for programmable logic controller. A PLC is a digital computer that is used for automation of industrial electromechanical processes, such as controlling different machines.

Here we use plc for controlling our inverter, sensor and other monitoring system.

Supervisory control and data acquisition: (SCADA)

Supervisory control and data acquisition (SCADA) is a system of software and hardware elements that allows industrial organizations to Control industrial processes locally or at remote locations, Monitor, gather, and process real-time data.

SCADA is directly interact with devices such as sensors, valves, pumps, motors, and more through human-machine interface (HMI) software.

SCADA systems are crucial for industrial organizations since they help to maintain efficiency, process data for smarter decisions, and communicate system issues to help mitigate downtime.

RS-485 MODBUS:

RS-485 allows multiple devices (up to 32) to communicate at half-duplex on a single pair of wires, plus a ground wire (more on that later), at distances up to 1200 meters (4000 feet). "Four-wire" networks have one master port with the transmitter connected to each of the "slave" receivers on one twisted pair.

IOT GATEWAY:

The IoT Gateway follows the same principle of bridging communications for different technologies. It creates a bridge between the sensors/actuators and the Internet. The IoT gateway aggregates all data obtained from communication port of digitizers and pre-processes the data along with security packets and converts to

internet compatibility protocols like GPRS or Ethernet or Wi-Fi and sends to database servers available either on cloud or locally.

IOT PLATFORM:

All the data collected from the above parameters are integrated to an IoT platform deployed on local or a cloud server. The IoT platform translates the data to information like graph, trends, alerts and other business intelligence reports.

BILL OF MATERIALS (BOM) PREPARATION:

S. N	Device/Instrument	Make	Model	Qty	Est unit price	Total
1.	TEMPERATURE SENSOR	MESSKO	MT-ST160F2	1	3200	3200
2.	DIGITIZER	KEYSIGHT	M8296A	1	6800	6800
3.	IOT GATEWAY	HUAWEI	RS-485	1	21000	21000
4.	PLC	SIEMENS	S7-300	1	30000	30000
5.	SCADA SOFTWARE	SOFTPRO	-	1	7400	7400
6,	SERVER & IOT PLATFORM	MICROSOFT AZURE	-	1 yr	32000	32000
7.	PANEL BOX	WORKSHOP	400 * 400	1	3500	3500
	TOTAL					1,03,900

PROJECT OUTCOME:

- Understanding the problem statements in the solar power plant.
- Finding out the best solutions.
- Wire connection and verification.
- Components Exploration.
- IOT Communication Working.